

23.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 31



At a room temperature of $300K$, calculate the thermal noise generated by two resistors of $10K\Omega$ and $20K\Omega$ when the bandwidth is $10KHz$.

- A. $1.2868 \times 10^{-6}V, 1.819 \times 10^{-6}V$ B. $6.08 \times 10^{-6}V, 15.77 \times 10^{-6}V$
C. $16.66 \times 10^{-6}V, 2.356 \times 10^{-6}V$ D. $1.66 \times 10^{-6}V, 0.23 \times 10^{-6}V$

nielit2017july-scientistb-cs non-gatecse

Answer key

23.0.2 NIELIT 2016 DEC Scientist B (CS) - Section B: 60



The sequence of operation in which PCM is done is

- A. Sampling, quantizing, encoding B. Quantizing, sampling, encoding
C. Quantizing, encoding, sampling D. None of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key

23.0.3 NIELIT 2016 DEC Scientist B (CS) - Section B: 41



The IETF standard documents are called:

- A. RFC B. RCF C. ID D. none of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key

23.0.4 NIELIT 2016 DEC Scientist B (CS) - Section B: 25



A low pass filter is

- A. Passes the frequencies lower than the specified cut off frequency
B. Used to recover signal from sampled signal
C. Rejects higher frequencies
D. All of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key

23.0.5 NIELIT 2016 DEC Scientist B (CS) - Section B: 17



$T1$ carrier system is used:

- A. For delta modulation B. Industrial noise
C. For frequency modulated signals D. None of the above

nielit2016dec-scientistb-cs non-gatecse

Answer key

23.0.6 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 27



At 100% modulation, the power in each sideband is _____ of that of carrier.

- A. 50% B. 40% C. 60% D. 25%

nielit2017oct-assistanta-cs non-gatecse

23.0.7 NIELIT 2016 MAR Scientist C - Section C: 43



The astable multivibrator has

- A. two quasi stable states B. two stable states
C. one stable and one quasi-stable state D. none of these

nielit2016mar-scientistc non-gatecse

23.0.8 NIELIT 2016 MAR Scientist C - Section C: 42



A stable multivibrator are used as

- A. comparator circuit B. squaring circuit
C. frequency to voltage converter D. voltage to frequency converter

nielit2016mar-scientistc non-gatecse

23.0.9 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 12



The open-loop transfer function of a feedback control system is $G(s) \cdot H(s) = 1/(s + 1)^3$. The gain margin of the system is

- A. 2 B. 4 C. 8 D. 16

nielit2017oct-assistanta-cs non-gatecse

23.0.10 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 11



In MOSFET fabrication, the channel; length is defined during the process of

- A. Isolation oxide growth B. Channel stop implantation
C. Poly-silicon gate patterning D. Lithography step leading to the contact pad

nielit2017oct-assistanta-cs non-gatecse

23.0.11 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 8



A carrier $Ac \cos(\omega c)t$ is frequency modulated by a signal $Em \cos(\omega m)t$. The modulation index is mf . The expression for the resulting FM signal is

- A. $Ac \cos[\omega c t + mf \sin(\omega m)t]$
B. $Ac \cos[\omega c t + mf \cos(\omega m)t]$
C. $Ac \cos[\omega c t + \pi mf \sin \omega m t]$

D. $Ac \cos[\omega ct + 2\pi mfEm \cos(\omega m)t/\omega m]$

nielit2017oct-assistanta-cs non-gatecse

Answer key 

23.0.12 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 7



If the number of bits per sample in a PCM system is increased from a n to $n + 1$, the improvement in signal to quantization noise ratio will be

- A. 3 dB B. 6 dB C. $2n \text{ dB}$ D. $n \text{ dB}$

nielit2017oct-assistanta-cs non-gatecse

23.0.13 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 5



For a periodic signal $v(t) = 30 \sin 100t + 10 \cos 300t + 6 \sin(500t + \frac{\pi}{4})$, the fundamental frequency in rad/s

- A. 100 B. 300 C. 500 D. None of these

nielit2017oct-assistanta-cs non-gatecse

23.0.14 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 4



For the discrete signal $x[n] = a^n u[n]$, $a > 0$ the z -transform is

- A. $\frac{(z+a)}{z}$ B. $\frac{(z-a)}{z}$ C. $\frac{z}{(z-a)}$ D. $\frac{z}{(z+a)}$

nielit2017oct-assistanta-cs non-gatecse

23.0.15 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 3



The final value theorem is used to find the

- A. Steady state value of the system output B. Initial value of the system output
C. Transient behavior of the system output D. None of these

nielit2017oct-assistanta-cs non-gatecse

23.0.16 NIELIT 2017 OCT Scientific Assistant A (CS) - Section D: 2



An ideal op-amp is an ideal

- A. voltage controlled current source B. voltage controlled voltage source
C. current controlled current source D. current controlled voltage source

nielit2017oct-assistanta-cs non-gatecse

Answer key 

A. $\frac{5}{6} \Omega$ B. $\frac{1}{6} \Omega$ C. $\frac{6}{5} \Omega$ D. $\frac{3}{2} \Omega$



A. 16 kbps
B. 24 kbps
C. 32 kbps
D. 8 kbps



A. 50 B. 67 C. 150 D. 33



A. 1750 B. 2625 C. 4000 D. 5250



A. gives rise to aperture effect B. implies oversampling

C. lead to aliasing

D. introducing delay distortion

nielit2022apr-scientistb digital-signal-processing analytical-aptitude

Answer key 

23.3.2 Digital Signal Processing: NIELIT Scientific Assistant A 2020 November: 60

Term in the MVC architecture that receives events is called _____.



A. Receiver

B. Controller

C. Transmitter

D. Modulator

nielit-sta-2020 digital-signal-processing

Answer key 

23.4 Frequency Spectrum (1)

23.4.1 Frequency Spectrum: NIELIT 2022 April Scientist B | Section B | Question: 109

A 1 MHz sinusoidal carrier is amplitude modulated by a symmetrical square wave of period 100 μsec . Which of the following frequencies will not be present in the modulated signal ?



A. 990 MHz

B. 1010 MHz

C. 1020 MHz

D. 1030 MHz

nielit2022apr-scientistb analog-communication signal-processing frequency-spectrum

Answer key 

23.5 Microprocessors (2)

23.5.1 Microprocessors: NIELIT 2016 MAR Scientist C - Section C: 48

In a microprocessor, WAIT states are used to



- A. make the processor wait during a DMA operation
- B. make the processor wait during a power interrupt processing
- C. make the processor wait during a power shutdown
- D. interface slow peripherals to the processor

nielit2016mar-scientistc non-gatecse microprocessors

Answer key 

23.5.2 Microprocessors: NIELIT 2017 DEC Scientific Assistant A - Section B: 8

Microprocessors are used in which generation of computers?



A. I – st Generation

B. II – nd Generation

C. III – rd Generation

D. IV – th Generation

nielit2017dec-assistanta non-gatecse microprocessors

Answer key 

23.6

Microprogramming (1)

23.6.1 Microprogramming: NIELIT 2016 MAR Scientist C - Section C: 61



Microprogramming is a technique for

- A. writing small programs effectively
- B. programming output/input routines
- C. programming the microprocessors
- D. programming the control steps of a computer

nielit2016mar-scientistc co-and-architecture microprogramming

Answer key

23.7

Signal Processing (2)

23.7.1 Signal Processing: NIELIT 2021 Dec Scientist A - Section B: 80



A modulating signal $m(t) = 10 \cos(2\pi \times 10^3 t)$ is amplitude modulated with a carrier signal $c(t) = 50 \cos(2\pi \times 10^5 t)$. Assume $R = 1\Omega$. Find the carrier power required for transmitting this AM wave.

- A. 1000 W
- B. 1250 W
- C. 1100 W
- D. 50 W

nielit2021dec-scientista analog-communication signal-processing

23.7.2 Signal Processing: NIELIT 2021 Dec Scientist B - Section B: 44



$y = 10 \cos(1800 \pi t) + 20 \cos(2000 \pi t) + 10 \cos(220 \pi t)$. Find the modulation index (μ) of the given wave.

- A. 0.3
- B. 0.5
- C. 0.7
- D. 1

nielit2021dec-scientistb analog-communication signal-processing

23.8

Signals and Systems (1)

23.8.1 Signals and Systems: NIELIT 2021 Dec Scientist A - Section B: 64



Energy of power signal is :

- A. Finite
- B. Zero
- C. Infinite
- D. 1

nielit2021dec-scientista signals-and-systems analytical-aptitude

Answer key

Answer Keys

23.0.1	A	23.0.2	A	23.0.3	A	23.0.4	D	23.0.5	D
23.0.6	D	23.0.7	D	23.0.8	B	23.0.9	C	23.0.10	C
23.0.11	A	23.0.12	B	23.0.13	A	23.0.14	C	23.0.15	A

23.0.16	B
23.3.1	A
23.6.1	D

23.0.17	A
23.3.2	B
23.7.1	B

23.0.18	B
23.4.1	C
23.7.2	D

23.1.1	B
23.5.1	D
23.8.1	C

23.2.1	C
23.5.2	D